10.12 Lorentz Invariance of Cross Sections

The cross-section quantifies the likelihood of a scattering process (most commonly used with two incoming particles). A property of cross sections is their **Lorentz invariance**: they remain unchanged under Lorentz transformations (boosts, rotations). This short note explains why cross sections are Lorentz invariant, despite their dimensional interpretation as "areas," and clarifies a common misconception.

The Paradox of Cross Sections as Invariant "Areas"

Cross sections (σ) have dimensions of area ([Length]^2), leading to the apparent contradiction: *"Areas generally change under boosts; how can σ, defined as an area, be Lorentz invariant?"*
The resolution lies in distinguishing between:

- **Geometric areas**: Physical spatial regions subject to Lorentz contraction.
- **Cross sections**: Derived quantities in QFT constructed from Lorentz-invariant ingredients.

Cross sections are not geometric objects but **Lorentz scalars** built from invariant components:

$$\sigma = \frac{|\mathcal{M}|^2 \cdot (\text{LIPS})}{\text{Flux Factor}}, \tag{10.12.1}$$

where:

- $|\mathcal{M}|^2$: Lorentz-invariant scattering amplitude.
- LIPS: Lorentz-invariant phase space.
- Flux Factor: Lorentz-invariant collision flux.

For two colliding particles with four-momenta p_1, p_2,

$$\text{Flux Factor} = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}.$$

The flux factor is manifestly invariant, as it is written in terms of Lorentz invariant quantities. The "tricky" point here is that we motivated the flux factor from a particular frame and then "dressed it up" with normalization factors from $\langle i|i\rangle/V$ (which, for two particles, scaled as V).

Definition

Cross sections are defined as:

$$\sigma = \frac{\text{interaction rate per target particle}}{\text{incident flux (particles per unit area per unit time)}}. \tag{10.12.2}$$

A common misconception is that the flux is invariant because the relative velocity is. It is not: the relative velocity v_{rel} is frame-dependent, and the invariant flux is its relativistic *generalization*.

Experimental Consistency (Convenience)

Cross section invariance ensures experimental results are frame-independent:

- **Example**: The same scattering process measured in a collider (center-of-mass frame) and in a fixed-target lab frame, at the same $\sqrt{s}$, yields identical σ.
- **Importance**: Lorentz invariance is *required* for consistent predictions across all inertial frames.